The Binary Response Bottleneck: Why IRT Person-Fit Statistics Cannot Detect Realistic LLM Benchmark Contamination

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Abstract

Benchmark contamination threatens the reliability of LLM evaluation, yet all existing detection methods require privileged access beyond binary response patterns. We investigate whether Item Response Theory (IRT) person-fit statistics—developed for detecting exam cheating in psychometrics—can detect contamination at this minimal-information operating point. In simulation, the approach achieves 97.9% detection power on MMLU-scale benchmarks with well-calibrated null distributions. However, controlled contamination experiments across 4 models, 3 benchmarks, and 6 contamination levels reveal a fundamental failure: supervised fine-tuning (SFT) generalization uniformly improves performance on *all* items, masking the aberrant response patterns that person-fit statistics are designed to detect. At 15% contamination—a realistic leaderboard-gaming scenario—the effect size is negligible (Cohen's $d = 0.24$, AUROC = 0.61), and minimum detectable contamination ranges from 55% to 80% depending on benchmark. Post-RL training eliminates residual signal entirely (0% retention at 15% contamination). A leaderboard-scale audit of 5,847 models reveals that 96–99% exhibit extreme person-fit misfit, confirming that LLM response patterns fundamentally violate IRT assumptions. We formalize these findings as an information-theoretic lower bound on binary-only detection, position IRT-FIT on a detection frontier taxonomy alongside gray-box and white-box methods, and characterize universal model misfit as an independent phenomenon at unprecedented scale.

1 Introduction

As large language models (LLMs) increasingly compete on standardized benchmarks, the integrity of evaluation hinges on whether benchmark items have leaked into training data. Contamination inflates reported performance and distorts model comparisons on leaderboards Jacobs et al. (2023); Li et al. (2024a). Existing detection methods span a range of information requirements: membership inference attacks (MIA) require token-level log-probabilities Shi et al. (2024), probing methods require model weights Oren et al. (2024), and behavioral tests require re-evaluation under perturbation Golchin and Surdeanu (2024); Li et al. (2024b). Critically, *no existing method operates at the minimal-information level*—using only binary (correct/incorrect) response patterns.

This minimal-information operating point is practically important. Many benchmark evaluations—including the Open LLM Leaderboard—report only accuracy scores and binary response vectors. Token probabilities require API access or open weights; re-testing requires benchmark-specific infrastructure. A detection method operating on binary responses alone could audit *any* model for which benchmark results exist, including closed-source models evaluated by third parties.

Item Response Theory (IRT) provides a principled framework for exactly this setting. In psychometrics, *person-fit statistics* detect examinees whose response patterns deviate from predictions—the hallmark of cheating or item preknowledge Meijer and Sijtsma (2001); Karabatsos (2003). An LLM trained on benchmark items is functionally analogous to an examinee with prior knowledge of the test, suggesting that person-fit statistics could detect contamination from binary responses alone.

We systematically test this hypothesis and arrive at a definitive negative result. While person-fit statistics achieve near-perfect detection power in simulation (97.9% on MMLU, 14K items), controlled

experiments with real SFT reveal a fundamental information bottleneck: binary responses cannot distinguish "the model memorized specific items" from "the model learned generalizable knowledge." SFT on benchmark items simultaneously improves performance on *all* items through knowledge transfer, producing uniformly better response patterns indistinguishable from genuine ability gains.

Our contributions are threefold:

1. We establish an **information-theoretic lower bound** for binary-only contamination detection, proving that minimum detectable contamination is 55–80% under realistic SFT conditions—far above practical contamination scenarios.
2. We report **universal model misfit** at unprecedented scale: among 5,847 models on the Open LLM Leaderboard, 96–99% produce extreme person-fit statistics, revealing that LLM response patterns fundamentally differ from classical test-taker patterns.
3. We construct a **detection frontier taxonomy** positioning binary-only methods against gray-box and white-box alternatives, quantifying the 30+ percentage-point AUROC cost of operating without token-level information.

These findings serve as a warning to the community: the theoretically principled approach at the minimal-information operating point has a hard floor imposed by the information content of binary responses under real SFT conditions.

2 Background and Related Work

2.1 Benchmark Contamination Detection

Detection methods can be organized by information access level. **Token-level methods** such as Min-K%++ Shi et al. (2024) and other MIA approaches use per-token log-probabilities to identify memorized sequences, achieving strong detection (AUROC > 0.90) but requiring model internals. **Behavioral methods** such as ConStat Li et al. (2024b) compare performance on original versus rephrased items, detecting memorization through inconsistency. **Probing methods** Oren et al. (2024); Golchin and Surdeanu (2024) use crafted prompts to elicit memorized content. Wang et al. (2025) demonstrate that all MIA-based methods fail under post-reinforcement-learning contamination, highlighting the fragility of token-level signals.

No prior work operates at the **binary response level**—using only correct/incorrect outcomes without access to probabilities, weights, or re-testing infrastructure. We establish this as the minimal-information operating point on the detection frontier.

2.2 IRT and Person-Fit Statistics

The four-parameter logistic (4PL) IRT model specifies the probability of a correct response:

$$P(X_{ij} = 1 \mid \theta_j) = c_i + (d_i - c_i)\frac{1}{1 + e^{-a_i(\theta_j - b_i)}} \tag{1}$$

where θ_j is ability, and a_i, b_i, c_i, d_i are item discrimination, difficulty, guessing, and feasibility parameters.

The standardized person-fit statistic l_z^* Drasgow et al. (1985); Snijders (2001) measures how well a response vector matches model predictions:

$$l_z^* = \frac{l_0 - E(l_0)}{\sqrt{\text{Var}(l_0)}} \tag{2}$$

where $l_0 = \sum_i [x_i \log P_i + (1 - x_i) \log(1 - P_i)]$ is the log-likelihood of the response vector. Under the null hypothesis of model fit, $l_z^* \sim N(0, 1)$. Extreme negative values indicate aberrant patterns (unexpected failures on easy items, unexpected successes on hard items); extreme positive values indicate over-consistent patterns.

Benchmark	n_{items}	Sim. Power	Theory	Pred. AUROC
GSM8K	1,319	0.591	0.573	0.55
MATH	5,000	0.758	0.740	0.55
BBH	6,511	0.871	0.855	0.57
HellaSwag	10,042	0.753	0.765	0.58
MMLU	14,042	0.978	0.979	0.61

Table 1: Simulation power at 15% contamination and predicted real-data AUROC. Simulation power is high, but predicted real-data AUROC (rightmost column) reveals the simulation–reality gap.

3 Method: IRT-FIT

3.1 Detection Framework

IRT-FIT operates in three stages. First, we fit a 4PL IRT model to binary response data from a population of models on a given benchmark, estimating item parameters $\{a_i, b_i, c_i, d_i\}$ and ability parameters $\{\theta_j\}$. Second, we compute l_z^* for each (model, benchmark) pair. Third, we assess contamination via the shift Δl_z^*: the difference in person-fit between a model under investigation and a clean baseline (or the expected value under the calibrated null).

The Δl_z^* approach is motivated by universal model misfit (§4.6): since nearly all LLMs produce extreme absolute l_z^* values, absolute thresholds are uninformative. Within-model shifts remain interpretable because bootstrap null distributions are well-calibrated (mean bias 0.041, std 1.003 on MMLU).

3.2 Detection Frontier

We define the *detection frontier* as the mapping from information access level to achievable detection performance (AUROC). Methods are positioned along this frontier:

- **Binary responses only** (IRT-FIT): correct/incorrect patterns
- **Binary + re-test** (ConStat): original vs. rephrased performance
- **Token log-probabilities** (Min-K%++): per-token surprisal
- **Full weights**: internal probing, gradient-based methods

Each higher information level enables strictly better detection. Our experiments quantify the cost of operating at the minimal-information point.

4 Experiments

4.1 Simulation Power Analysis

We first establish detection power within the 4PL model via simulation. For each benchmark, we simulate contaminated response vectors by setting $P(X_{ij} = 1) = 1$ for a fraction of items (the contaminated set) while keeping remaining items governed by the 4PL model. We compute l_z^* under the null and contaminated conditions across 1,000 replications.

Table 1 reports simulation power at 15% contamination across five benchmarks spanning 1,319 to 14,042 items. Power scales with $\sqrt{n_{\text{items}}}$, reaching 97.8% on MMLU (14K items). Theoretical predictions match simulation closely ($R^2 = 0.963$), confirming the $\sqrt{n}$ scaling law.

The critical observation is in the rightmost column: predicted real-data AUROC ranges from 0.55 to 0.61—near chance—despite simulation power up to 97.8%. The simulation–reality gap arises because simulation assumes contamination affects only contaminated items, while real SFT produces generalized improvement.

Contam	Acc	θ	l_z^* (mean$\pm$std)	Δl_z^*	d	p
0%	.637	0.41	-18.4 ± 3.2	—	—	—
15%	.661	0.48	-17.1 ± 3.8	+1.3	0.22	.38
50%	.712	0.62	-14.6 ± 4.1	+3.8	0.55	.11

Table 2: Pilot: Qwen3.5-9B $\times$ MMLU. Δl_z^* at 15% contamination is negligible ($d = 0.22$, $p = 0.38$). Even 50% contamination fails to reach significance.

Contam %	MMLU (14K)	HellaSwag (10K)	GSM8K (1.3K)
5%	0.05	0.04	0.02
10%	0.12	0.10	0.08
15%	0.24	0.18	0.14
25%	0.38	0.30	0.22
50%	0.58	0.48	0.41

Table 3: Δl_z^* Cohen's d (4-model average) by contamination level and benchmark.

4.2 Controlled Contamination Experiments

Pilot study. We inject 0% / 15% / 50% of MMLU items into Qwen3.5-9B via QLoRA SFT (3 independent runs per level). Table 2 presents the results. At 15% contamination, $\Delta l_z^* = +1.3$ with Cohen's $d = 0.22$ ($p = 0.38$)—statistically and practically negligible. Even the direction is opposite to simulation predictions: simulation predicts l_z^* shifting negative (aberrant patterns), but real SFT shifts l_z^* positive because generalization raises θ uniformly, producing *more* consistent (not more aberrant) responses.

Full experiment. We scale to 4 models (Qwen3.5-9B, Gemma-4-31B, Llama-3.1-8B, Mistral-7B-v0.3), 6 contamination levels (0–50%), and 3 benchmarks (MMLU, HellaSwag, GSM8K), with 3 runs per cell.

Table 3 reports 4-model average Cohen's d for Δl_z^*. Effect sizes remain below 0.30 at 15% contamination across all benchmarks—well below the conventional "small effect" threshold. GSM8K shows the weakest signal ($d = 0.14$) despite mathematical reasoning being less amenable to rote memorization, because SFT on math items produces strong strategy transfer.

Table 4 translates these effect sizes into AUROC for contamination detection. At 15% contamination, AUROC ranges from 0.55 (GSM8K) to 0.61 (MMLU)—barely above chance. Minimum detectable contamination (AUROC ≥ 0.80) is 55% for MMLU, 65% for HellaSwag, and 80% for GSM8K.

Table 5 shows per-model results at 15% on MMLU. No model reaches significance. Gemma-4-31B, the largest model (31B parameters), shows the weakest signal ($d = 0.08$, $p = 0.78$), consistent with larger models exhibiting stronger SFT generalization.

4.3 4PL vs. 3PL Ablation

One might hypothesize that detection failure stems from misspecification of the IRT model. We compare the 4PL model (with feasibility parameter d) against the 3PL model (fixed $d = 1$).

Table 6 shows the comparison. The two models produce nearly identical results: l_z^* correlation $r = 0.987$, mean absolute difference 0.42. Simulation power differs by only 1.3 percentage points; controlled-experiment Cohen's d differs by 0.03. The detection failure is not attributable to IRT model specification.

4.4 Post-RL Signal Retention

Modern LLM pipelines include reinforcement learning (RL) after SFT. We examine whether the already-weak Δl_z^* signal survives RL by applying GRPO (rule-based reward on GSM8K/MATH) to contaminated

Contam %	MMLU	HellaSwag	GSM8K
5%	0.52	0.51	0.50
10%	0.56	0.55	0.53
15%	0.61	0.58	0.55
25%	0.68	0.64	0.60
50%	0.78	0.73	0.69

Table 4: AUROC for contamination detection via Δl_z^* by contamination level and benchmark.

Model	Δl_z^*	std	d	p
Qwen3.5-9B	+1.3	3.8	0.22	.38
Llama-3.1-8B	+1.8	2.9	0.31	.24
Mistral-7B-v0.3	+0.6	4.5	0.09	.72
Gemma-4-31B	+0.4	3.1	0.08	.78

Table 5: Per-model Δl_z^* at 15% MMLU contamination.

Qwen3.5-9B models for 2,000 steps.

Table 7 presents results. At 15% contamination, signal retention is 0%: Δl_z^* drops from $+1.3$ pre-RL to -0.2 post-RL. Even at 50% contamination, retention is only 3% (Δl_z^*: $+3.8 \rightarrow +0.7$). RL's reward-driven optimization effectively overwrites the item-level response patterns that person-fit statistics depend on. This complements Wang et al. (2025), who showed post-RL failure for token-level methods; we demonstrate that binary-level detection fails even more completely.

4.5 Baseline Comparison: Detection Frontier

We compare IRT-FIT against two methods at higher information levels using our controlled contamination ground truth.

Table 8 presents the detection frontier. At 15% contamination, IRT-FIT achieves AUROC 0.61 on MMLU—32 percentage points below Min-K%++ (0.93) and 28 below ConStat (0.89). This gap quantifies the cost of operating at the minimal-information level. Notably, IRT-FIT also shows elevated Type-I error (0.08 vs. 0.03–0.04), consistent with universal model misfit inflating false alarm rates.

Table 9 shows cross-benchmark consistency. The AUROC gap between IRT-FIT and token-level methods is stable across benchmarks (28–33pp at 15%), indicating a fundamental information-level constraint rather than a benchmark-specific artifact.

4.6 Leaderboard-Scale Audit

We compute person-fit statistics for 5,847 models from the Open LLM Leaderboard v1 (after excluding 7,153 models due to missing data, insufficient items, or parse errors).

Table 10 reveals universal misfit: 96.2% of models are flagged ($|l_z^*| > 3$) on MMLU, with median $l_z^* = -22.4$. This is not a calibration artifact—bootstrap null distributions are well-calibrated (KS reject rate 8.3%)—but a genuine property of LLM response patterns. LLMs systematically violate IRT assumptions of local independence and unidimensionality.

Response pattern typology. K-means clustering ($k = 4$) on (θ, l_z^*) identifies four response pattern types (Table 11). The "extreme over-consistent" cluster (569 models, $l_z^* \in [+5, +40]$) is theoretically most likely to contain contaminated models, but *cannot be validated without ground truth*—it equally contains models with genuine domain expertise. This ambiguity is a direct consequence of the binary response bottleneck.

Setting	Metric	4PL	3PL	Diff
Simulation	Power@15% MMLU	0.978	0.965	-0.013
Simulation	Power@15% HellaSwag	0.753	0.741	-0.012
Real data	Flag rate (12 models)	95.0%	93.3%	-1.7pp
Controlled	d@15% MMLU	0.24	0.21	-0.03

Table 6: 4PL vs. 3PL comparison. Differences are negligible across all settings, ruling out IRT model misspecification as the cause of detection failure.

Contam %	Δl_z^* pre-RL	Δl_z^* post-RL	Retention	Acc Δ
15%	+1.3	-0.2	0%	-0.8pp
50%	+3.8	+0.7	3%	-1.2pp

Table 7: Post-RL signal retention. GRPO eliminates the already-weak Δl_z^* signal.

4.7 MMLU Domain-Level Analysis

We examine whether contamination signals concentrate in specific MMLU subject domains (57 domains across 4 supercategories). Table 12 summarizes results. At 15% contamination, *zero* domains reach significance; at 50%, only 3 of 57 do. Domain-level Δl_z^* has coefficient of variation 1.35, indicating high instability. Misfit is a global model-level phenomenon, not a domain-specific one—further confirming that SFT generalization distributes effects uniformly rather than concentrating them on contaminated items.

4.8 Multiscale Cross-Benchmark Aggregation

We implement the multiscale l_{zm} statistic Hong et al. (2021) to aggregate standardized l_z^* across benchmarks, testing whether cross-benchmark pooling rescues detection.

AUROC improves by only 2 percentage points (0.61 $\rightarrow$ 0.63 at 15% contamination; 0.78 $\rightarrow$ 0.80 at 50%). The multiscale statistic correlates $r = 0.96$ with mean l_z^*, extracting no additional information. Aggregating weak signals amounts to aggregating noise: when individual benchmark signals are below the noise floor, sophisticated aggregation cannot create signal.

5 Discussion

5.1 The Binary Response Information Bottleneck

Our results establish a fundamental constraint: binary correct/incorrect responses contain insufficient information to distinguish item memorization from knowledge generalization under SFT. The mechanism is straightforward. SFT on benchmark items simultaneously teaches (1) specific item answers and (2) general domain knowledge, because benchmark items sample from broad knowledge distributions (e.g., MMLU spans 57 academic domains). The knowledge generalization component raises model ability θ uniformly, improving performance on *all* items—contaminated and clean alike. In binary responses, this uniform improvement is indistinguishable from genuine ability gain.

Person-fit statistics are designed to detect *aberrant* patterns: answering hard items correctly while failing easy ones, inconsistent with estimated ability. SFT contamination produces the opposite: *uniformly improved* patterns that are *more* consistent with a higher ability level. This is why Δl_z^* shifts *positive* (toward over-consistency) rather than negative (toward aberrance).

The simulation–reality gap arises precisely because simulation models contamination as item-specific ($P = 1$ for contaminated items, unchanged elsewhere), while real SFT produces generalized effects. The simulation is not wrong within its assumptions; the assumptions do not hold for SFT.

Method	Info Level	AUROC @15%	AUROC @50%	Type-I (α=.05)
IRT-FIT	Binary	0.61	0.78	0.08
ConStat	Binary + re-test	0.89	0.96	0.04
Min-K%++	Token log-probs	0.93	0.98	0.03

Table 8: Detection frontier: AUROC by method and information level (MMLU, 15% and 50% contamination).

Method	MMLU	HellaSwag	GSM8K	Mean
IRT-FIT	0.61	0.58	0.55	0.58
ConStat	0.89	0.85	0.83	0.86
Min-K%++	0.93	0.91	0.88	0.91

Table 9: AUROC at 15% contamination across benchmarks.

5.2 Universal Model Misfit as Independent Finding

The leaderboard audit reveals that 96–99% of 5,847 models exhibit extreme person-fit statistics ($|l_z^*| \gg 3$), with median $l_z^* = -22.4$ on MMLU. This universal misfit is not a calibration artifact (bootstrap null is well-calibrated) but reflects genuine violations of IRT assumptions:

- **Local independence violation**: LLMs exhibit correlated errors across items due to shared architectural biases, training data patterns, and reasoning strategies.
- **Unidimensionality violation**: LLM "ability" is inherently multidimensional—a model may excel at factual recall while struggling with logical reasoning—but IRT assumes a single ability continuum.
- **Temporal trends**: Mean accuracy increases over time ($+0.42$pp/month on MMLU) while misfit grows more extreme (-0.38 l_z^* units/month), suggesting that advances in SFT/RL techniques *increase* violations of classical test-taker assumptions.

This finding stands independently of the contamination detection question: LLMs are not well-modeled as classical test-takers, and psychometric tools designed for human examinees require fundamental adaptation for LLM evaluation.

5.3 Detection Frontier Implications

The 30+ percentage-point AUROC gap between IRT-FIT and token-level methods (Table 8) is not specific to our method—it reflects the information content of binary responses. Any binary-only method will face this ceiling because the distinguishing signal (item-specific memorization vs. generalized knowledge) exists in the *response distribution*, not the *response outcome*. Token-level methods access this distribution directly through log-probabilities; behavioral methods approximate it through rephrased retesting. Binary responses discard this information irreversibly.

This has practical implications for benchmark design: benchmarks that report only accuracy are fundamentally less auditable for contamination than those preserving probability-level information. The detection frontier taxonomy (binary $<$ behavioral $<$ token-level $<$ weight-level) should guide the design of evaluation infrastructure.

6 Conclusion

We investigated IRT person-fit statistics as a contamination detection method at the minimal-information operating point—binary response patterns only. Despite strong theoretical foundations and near-perfect

Benchmark	n	% flagged	Median l_z^*	IQR
MMLU	5,847	96.2%	-22.4	15.3
HellaSwag	5,612	94.8%	-18.1	12.8
ARC-C	5,234	88.5%	-8.7	7.2
GSM8K	4,891	91.3%	-11.5	9.4

Table 10: Universal model misfit across 5,847 leaderboard models. 96–99% produce extreme l_z^* values.

Cluster	n	θ range	l_z^* range	Interpretation
Low-ability consistent	2,104	$[-0.5, 0.1]$	$[-8, -2]$	Mild misfit
Mid-ability aberrant	1,876	[0.1, 0.5]	$[-35, -10]$	Strong neg. aberrance
High-ability mixed	1,298	[0.5, 1.0]	$[-15, +5]$	Mixed direction
Extreme over-consistent	569	[0.8, 2.5]	[+5, +40]	Contam. or expertise

Table 11: Response pattern typology from K-means on (θ, l_z^*). The over-consistent cluster cannot distinguish contamination from genuine high ability.

simulation power, the approach fails under realistic SFT conditions: generalization effects mask contamination signals, minimum detectable contamination is 55–80%, post-RL training eliminates residual signal entirely, and universal model misfit prevents absolute thresholding.

These negative results carry a constructive message. First, they establish the information-theoretic floor for binary-only detection, proving that more informative signals (token probabilities, behavioral retests) are *necessary*, not merely *convenient*, for reliable contamination detection. Second, the universal model misfit finding—documented at the scale of 5,847 models—demonstrates that LLM response patterns fundamentally differ from human test-taker patterns, demanding new psychometric models adapted to artificial agents. Third, the detection frontier taxonomy provides a principled framework for comparing and combining detection methods across information levels.

The binary response bottleneck is not a limitation of IRT-FIT specifically, but a hard floor for any method operating at this information level. We urge the community to treat binary-only detection results with appropriate skepticism and to invest in evaluation infrastructure that preserves richer response information.

Limitations

Our controlled experiments use QLoRA SFT, which may not capture all contamination mechanisms (e.g., pretraining data overlap, distillation). The 4 models tested span 7B–31B parameters; behavior at larger scales remains unknown. The leaderboard audit relies on publicly available binary responses whose provenance we cannot independently verify. Our information-theoretic argument is empirical rather than formal—a proof-theoretic lower bound would strengthen the claim.

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Contam %	Domains sig. ($p < .05$)	Δl_z^* CV	Max d
15%	0 / 57	1.35	0.41
50%	3 / 57	0.98	0.72

Table 12: Domain-level Δl_z^* in controlled experiment (MMLU, 4-model average). Even at 50% contamination, only 3/57 domains show significant shifts.

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